

Machine Learning assisted Structural Design Optimization for Flip Chip Packages

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Abstract—The development of fine-linewidth semiconductor manufacturing process imposes additional requirements on the design optimization. This paper proposes and validates a simulation driven design methodology for structural design optimization of chip package integration. Finite Element Analysis method is employed to perform stress simulation for chip packages and then serves as a training dataset generator for machine learning model development. The effects of chip design parameters on the first principal stress are studied. Multiple machine learning algorithms are applied and evaluated as surrogate models for stress prediction. The random forest algorithm is identified to have the best capability to perform stress prediction for chip package integration design.

Keywords—simulation driven design, finite element analysis, machine learning, design optimization

I. INTRODUCTION

Semiconductor industry has managed to achieve an exponential increase of transistor density in microprocessors, which leads to fruitful progress in the development of cloud computing, advanced manufacturing, and Internet of Things [1, 2]. Taiwan Semiconductor Manufacturing Company (TSMC) newly developed 3nm process chips have reached the density of 250 million per mm². At the same time, the number of input and output ports of electronic products is also increasing dramatically, which accelerates the development of light, thin, and high-integrated microprocessors. In order to meet the increasing electrical and mechanical demands on electronic chips, the flip chip packaging technology developed by IBM has been widely applied in the electronic packaging industry. This technology refers to depositing convex solder joints on the aluminum bonding block of the chip, and then connecting the chip and the conductive bumps on the substrate to the external circuit. It eliminates lead connection, and the chip is facing down to achieve the shortest circuit connection. Compared with WB (wire bonding) and TAB (tape automatic bonding), flip chip can achieve higher package density, smaller package size, better electrical and thermodynamic characteristics, and lighter weight.

However, flip chip technology still has some reliability problems. Solder bump is identified as the weakest part of the whole package. The miniaturization of components makes current and temperature aggregation becomes severe. The package is composed of various materials, and the mismatch of thermal expansion coefficients among the materials is the main cause of the solder joint failure. Due to the requirement of environmental protection, the solder bump is required to be lead-free, which results in an increase in the melting point of the

solder bump, and thus a higher temperature is required in the reflow process, causing greater stress in the connection process.

Based on the discussion above, researchers in the electronic packaging field have been conducted a large amount of research [3, 4] studying the failure mechanism, reliability improvement and design optimization. By applying finite element analysis (FEA) techniques, researchers have obtained a good understanding on effect of material properties on reliability [5]. While due to the cost of experiment and limitation of computation capability, a systematic methodology for chip package design optimization has not been fully established. And this blocks chip designers from effectively identifying the optimal design parameters for their products.

The purpose of this paper is to employ finite element analysis (FEA) and machine learning algorithms to establish a methodology to optimize chip package design based on a systematic analysis of the stress distribution in a chip package in the reflow process. An FEA model was first developed and employed to study the effects of key design parameters on stress distribution. Then this FEA model serves as a training data generator to train machine learning models for chip reliability prediction. Multiple machine learning algorithms are then developed and evaluated to perform design optimization based on the FEA simulations.

II. MACHINE LEARNING ALGORITHMS

In this study, multiple algorithms are employed to perform stress prediction for a specific chip package design. The prediction performance is evaluated by comparing with the FEA simulation results, and the best algorithm for stress prediction is identified for further design optimization purposes.

A. Ordinary Least Squares (OLS) Linear Regression

Linear regression models are a good starting point for establishing supervised learning, and the least-squares regression model is one of the most commonly used statistical algorithms in data science.

The goal of OLS linear regression is to obtain model parameters by reducing the gap between the true value of the response variable and the predicted value, that is, to minimize the residual sum of squares (RSS) from each point in the best fitting curve to the actual value. RSS is expressed in (1),

$$RSS = \sum_{t=1}^T (y_t - \hat{y}_t)^2 = \sum_{t=1}^T (y_t - \hat{\alpha} - \hat{\beta}x_t)^2 \quad (1)$$

where y_t is the actual value, $\hat{y}_t$ is the predicted value in the fitted curve, and $\hat{\alpha}$, $\hat{\beta}$ are the regression coefficients that need to be determined.

The OLS model can also be extended to the case of multiple independent variables. It simulates the correlation between dependent variables and arguments, as expressed in (2)

$$Y = a_0 + a_1X_1 + a_2X_2 + \dots + a_iX_i + E \quad (2)$$

where X_i represents the set of independent variables, a_i represents the parameter coefficient of the regression and E is the random error term. Its coefficients are estimated by least squares.

B. Least Absolute Shrinkage and Selection Operator (Lasso) Regression

OLS gives nonzero estimates to all coefficients, while Lasso regression [6] obtains a more refined model by constructing a penalty function to compress some coefficients and set some coefficients to 0. Lasso estimates can be defined in (3)

$$\hat{\beta}(\text{lasso}) = \arg \min \|y - \sum_{j=1}^p x_j \beta_j\|^2 + \lambda \sum_{j=1}^p |\beta_j| \quad (3)$$

where λ is a nonnegative regularization parameter. The second term in eq (3) is the so-called ℓ_1 penalty, which helps to reduce the risk of overfitting and plays the role of feature selection, which is very important to the success of Lasso. By solving the minimum value, the predicted value of β_j is obtained by model learning.

C. Ridge Regression

Ridge regression [7] is a biased estimation regression method specially used for collinear data analysis. In essence, it is an improved least squares estimation method. By abandoning the unbiased nature of the least squares method, the regression coefficients are obtained at the cost of losing some information and reducing accuracy.

The model supposes a data set contains n pieces of information, and each information contains M parameters to form a matrix X . And y is the matrix composed of parameters to be studied. The loss function is expressed in (4) and (5)

$$L(\theta) = \|X\theta - y\|^2 + \|\Gamma\theta\|^2 \quad (4)$$

$$\theta(\alpha) = (X^T X + \alpha I)^{-1} X^T y \quad (5)$$

With the increase of α , the absolute values of each element $\theta(\alpha)$ tend to become smaller and their deviation from the correct value is also larger. The trajectory of $\theta(\alpha)$ changing with the change of α is called ridge trace, and α can be determined by observing the stability of ridge trace. Ridge regression is a supplement to the least squares regression, which loses unbiasedness in exchange for high numerical stability, so as to obtain high calculation accuracy.

D. Random Forests (RF)

Random forest is a statistical learning algorithm [8]. It uses the method of repeated sampling to extract multiple samples from the original samples. For each sample, it carries out

decision tree modeling, combines multiple decision trees, and votes to obtain the final prediction result.

The output of the tree predictor $h(x, y)$ is numeric, where y is a random vector, assuming that I and J are two vectors independent of the training set. The mean square generalization error can be expressed by (6)

$$E_{I,J}(J - h(I))^2 \quad (6)$$

The random forest algorithm can be expressed by (7)

$$R(i_m, j_{m,n}) = \frac{n_{\text{left}}}{n} H(I_{\text{left}}) + \frac{n_{\text{right}}}{n} H(I_{\text{right}}) \quad (7)$$

where n_{left} is the number of trained samples on the left decision tree after segmentation, and n_{right} is the number of trained samples on the right. The H function is used to measure the amount of impurities contained in the node. I_{left} and I_{right} are the actual values of the trained samples on the left and right decision trees.

E. Support Vector Machine (SVM)

SVM is also based on statistical learning theory with the main principle to find an optimal solution hyperplane that meets the classification requirements, so that the blank area on both sides is large enough. The problem of constructing the optimal hyperplane is transformed into finding the equation (8) under constraints

$$\min \phi(w) = \frac{1}{2} \|w\|^2 = \frac{1}{2} (w^T * w) \quad (8)$$

The support vector machine algorithm can be expressed (9)

$$f(m) = \text{sgn}(w \cdot \phi(m) + b) = \text{sgn}(\sum_{i=1}^l a_i n_i \phi(m_i) \cdot \phi(m) + b) \quad (9)$$

In (9), $\phi(m)$ represents the kernel function, which has many forms, which greatly affects the accuracy of machine learning prediction and regression results. In this formula, both the objective function and the decision function only involve the inner product operation between the training samples, which avoids the complex high-dimensional operation in the high-dimensional space and only needs the inner product operation. However, the algorithm is particularly sensitive to vector changes and is only suitable for small-scale low latitude data sets.

F. K-nearest Neighbors Regression

K-nearest-neighbors (KNN) is an algorithm based on nonparametric method statistics [9]. The input to the algorithm is the data containing the k closest training examples. If it is for a regression problem, the output is the average of k nearest neighbors, whereas for a classification problem, the output is a class member.

The KNN algorithm can be expressed by (10)

$$y_t = \frac{1}{\sum_{s=1}^n E_{st}} \sum_{t=1}^n E_{st} y_s^t \quad (10)$$

where y_t represents the class prediction result of the monitored point in the test data set. y_s refers to the set of variables closest to the t -th target prediction point in the training data set, n is the value of k to determine. The algorithm determines the k data closest to the unknown sample. k is determined by E_{st} , which

determines the range and number of nearest neighbors as expressed in (11)

$$E_{st} = \frac{1}{\sqrt{\sum_{t=1}^n (z_t - y_t)^2}} \quad (11)$$

where z_t and y_t are the target point and reference data point. If $k=1$, the unknown sample point can be classified as the data point closest to it. In this study, we need to predict the unique output value of each state. When the class label is constantly changing according to the weight coefficient E_{st} , this method is converted to a regression method.

G. Evaluation Methods

There are four commonly used performance measures for machine learning models [10], including mean absolute error (MAE), mean square error (MSE), mean root mean square error (RMSE), coefficient of determination (R^2) to evaluate the prediction capability of the model, as expressed in eq (12) to eq (15)

$$MAE = \frac{1}{n} \sum_{i=1}^n |f(i) - y| \quad (12)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n |f(i) - y|^2 \quad (13)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n |f(i) - y|^2} \quad (14)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n |y(i) - f(i)|^2}{\sum_{i=1}^n y(i) - \bar{y}} \quad (15)$$

where $f(i)$ is the i -th predicted value, $y(i)$ is the i -th actual value, $\bar{y}$ is the average value, and n is the number of samples.

III. FINITE ELEMENT ANALYSIS OF STRESS DISTRIBUTION

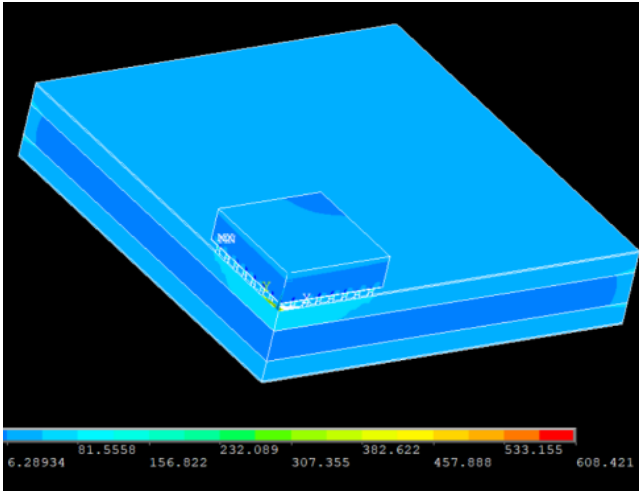


Fig. 1. Geometry of the chip package model with quarter symmetry.

We use ANSYS APDL to analyze the stress distribution of the chip in the reflow assembly process. Fig. 1 shows the model geometry of a microchip and the corresponding Printed Circuit Board (PCB) with quarter symmetry. When APDL is employed to build a general model, the geometric parameters of the model can be modified in a large range without rebuilding the model from scratch. Initially, chip thickness is 0.4 mm, length and width are 2.834 mm, PCB board is divided into 4 layers, total

thickness is 0.947 mm, length and width are 10 mm. Number of solder bumps is 64 and solder balls are mainly composed of copper, nickel and intermetallic compound (IMC).

A. FEA Simulation Results

We studied the effects of chip thickness, chip area and PCB dimension on the 1st principal stress of the chip packaged based on the established FEA model. Parameter z_{die} refers to the thickness of the chip. Parameter x_{die} and y_{die} refer to the length and width of the chip, and these two values are linked and set to be identical. Parameter x_{sub} and y_{sub} refer to the length and width of the Printed Circuit Board. Because the board is in square shape, the above two variables are always equal. Parameter z_{brd} is the total thickness of the PCB. By varying only one of the variables and restricting the other variables to remain unchanged, the effects of design parameters on the maximum 1st principal stress in the model were studied. It can be found from Fig. 1 that the maximum 1st principal stress occurred on the solder feet at the corner of the chip. This is in agreement with the experimental observation. Generally, the outermost welding foot is prone to breakage.

Fig. 2 reveals the trend of the first principal stress with respect to the thickness of the chip. The x-axis represents the thickness of the chip, with an interval of 0.05mm. And the y-axis refers to the maximum principal stress. In the range of z_{die} from 0 to 1mm, the first principal stress between chip and PCB increases gradually with the increase of chip thickness.

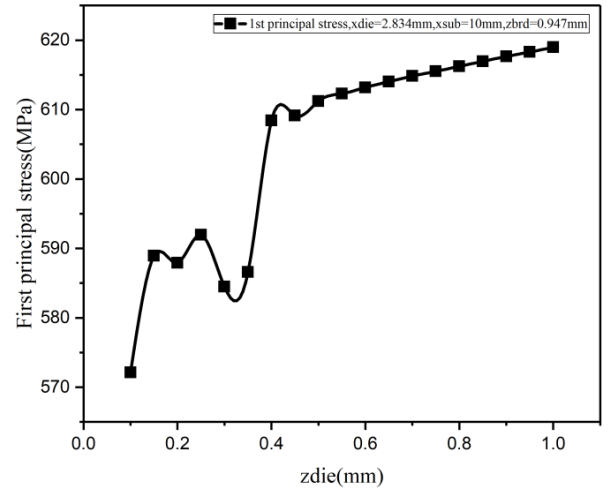


Fig. 2. The maximum 1st principal stress vs thickness of the chip.

Fig. 3 shows the relationship between the first principal stress and the width of the substrate board, where the x-axis is the width of the substrate and the interval between them is 0.05 mm. The figure shows that the first principal stress increases first and then decreases in the range of x_{die} from 2.734 to 3.534 mm. The stress is maximized when x_{die} is 2.784 mm. This is because when the chip is relatively small, the proportion relative to the PCB board is also small, which resulting in the mismatch issue and stress concentration at chip corner. When chip size becomes relatively large, the stress is still existing but distributes more evenly, which resulting in a diminishment of the maximum 1st principal stress in the structure.

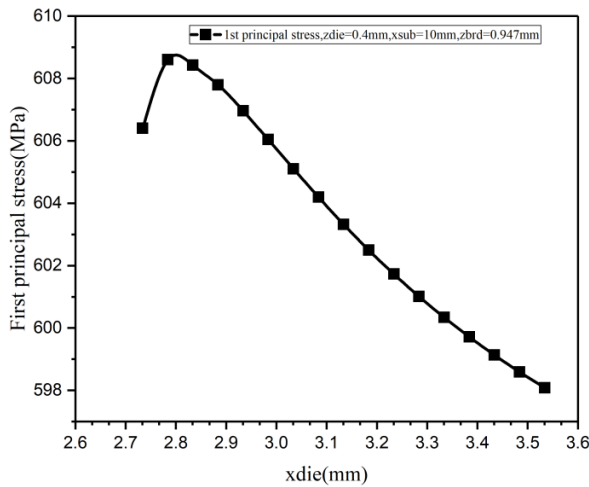


Fig. 3. The maximum 1st principal stress vs width of the chip.

Fig. 4 shows the effect of PCB board thickness on the first principal stress, where the x-axis is the thickness of the substrate and the interval between them is 0.05 mm. The figure shows that in the range of $zbrd$ from 0.897 to 1.697 mm, the first principal stress increases with the increasing PCB thickness. This is also because of the mismatch issue. When the thickness of PCB board is comparable to the chip thickness, the stress generated by the mismatch is relatively low. When the PCB thickness increases, more mismatch occurs which leads to a severe stress concentration.

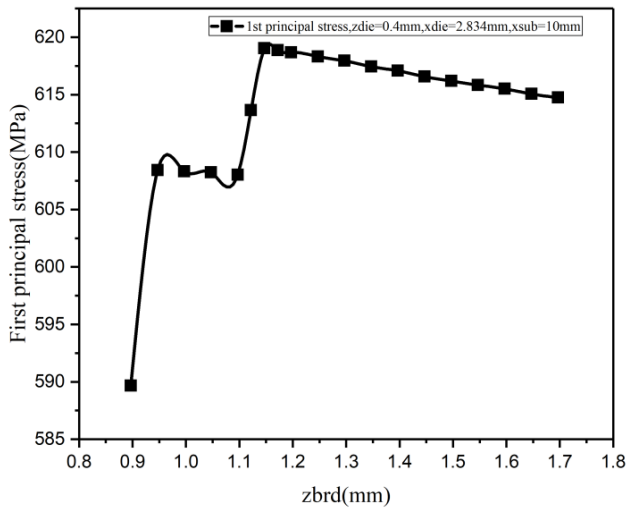


Fig. 4. The maximum 1st principal stress vs thickness of the PCB.

IV. MACHINE LEARNING MODEL FOR CHIP PACKAGE DESIGN OPTIMIZATION

A. Data Source and Pre-processing

Based on the developed FEA model, we varied the chip thickness, width, and PCB board thickness, and obtained the corresponding maximum first principal stress as the training data. Fig. 5 is the dataset distribution diagram. We varied only one variable at a time to generate the input parameter sets for

training purpose. The parameter name, its corresponding sample ID, and varying range for model training are listed in Table I.

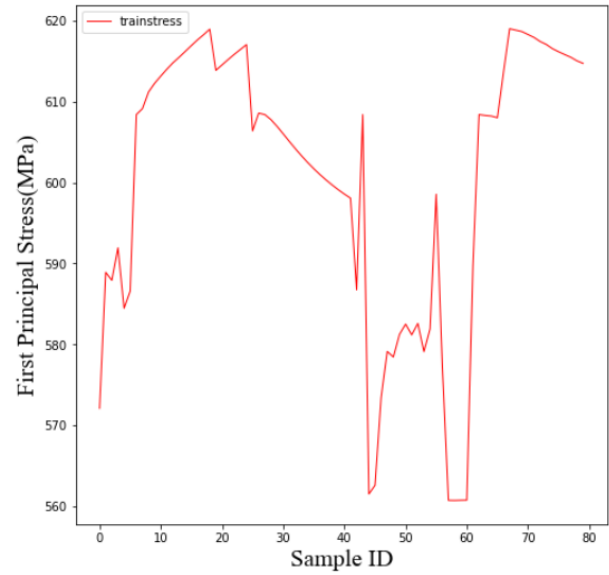


Fig. 5. Training dataset generated from the FEA model.

We also calculated the correlation between the independent variable and the dependent variable to ensure the accuracy of the prediction. The results show that all the four variables have a strong correlation. Then we set the first principal stress to be a dependent variable in the learning model and used several algorithms to predict the value of the first principal stress based on given design parameters including $zdie$, $xdie$, $xsub$ and $zbrd$. The following machine learning algorithms were used to perform stress prediction including ordinary least squares linear regression, lasso regression, ridge regression, poisson regression, bagging regression, SVR, KNN, and RF.

TABLE I. PARAMETER RANGE AND CORRESPONDING SAMPLE ID

Parameter	Sample ID	Varying Range
Chip Thickness	0 - 24	0.1 - 1.3 mm
Chip width	25 - 41	2.73 - 3.53 mm
PCB board width	42 - 60	6 - 95 mm
PCB board thickness	61 - 79	0.89 - 1.697 mm

B. Performance of Machine Learning Models

The prediction results from different algorithms are shown in Fig. 6, in which the first 25 data are selected for display, which is represented by sample id. The blue line in Fig. 6 represents the predicted first principal stress value, whereas the red line represents the actual value. It can be seen that the predicted results of Bagging, KNN, and RF are matched well with the test values compared to the linear regression.

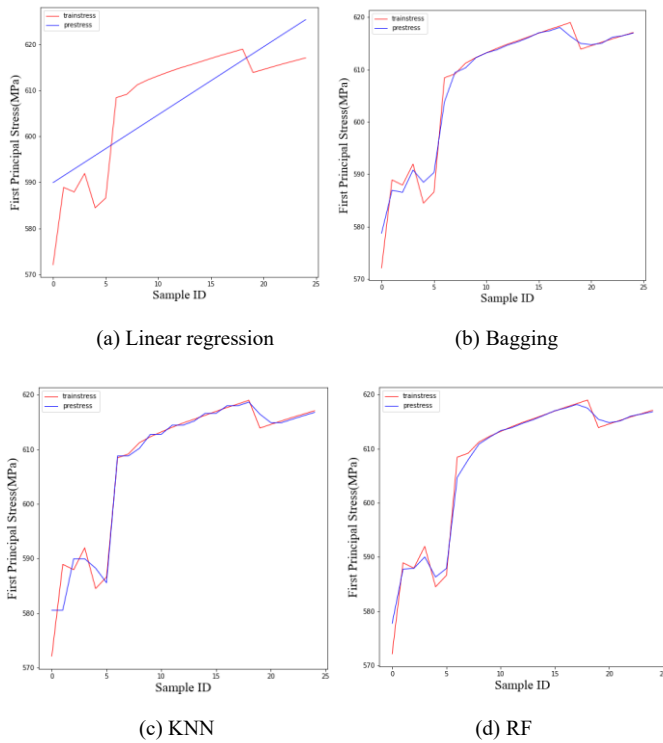


Fig. 6. Prediction performance of machine learning models.

In order to evaluate the performance of the algorithm in a more quantitative manner, we calculated the mean absolute error (MAE), mean square error (MSE), mean root mean square error (RMSE), and coefficient of determination (R^2) respectively for each algorithm. The results are shown in Table II.

TABLE II. COMPARISON OF PREDICTION ACCURACY OF DIFFERENT ALGORITHMS

Algorithm	Inspection method			
	R^2	MAE	MSE	RMSE
Linear Regression	0.701	7.144	89.616	9.467
Lasso	0.701	7.121	89.630	9.467
Ridge	0.664	6.979	100.789	10.039
Poisson Regression	0.700	7.106	90.044	9.489
SVR	0.557	7.184	132.810	11.524
KNN Regression	0.948	1.740	15.672	3.959
Bagging Regression	0.973	1.143	8.156	2.856
Random Forest	0.976	1.122	7.335	2.708

By comparing four evaluation criteria in Table I, it can be found that the KNN, Bagging and RF algorithms perform significantly better than other linear algorithms and SVR algorithms. Among all evaluated algorithms, the random forest algorithm has the best fitting performance.

V. CONCLUSION

In this paper, we used Finite Element Analysis to model the chip package integration. By investigating the effects of chip and PCB dimensions on the maximum first principal stress, we identified the optimal design parameters for chip package integration. We also proposed and established a machine learning model to predict the maximum first principal stress in a specific chip package design. This machine learning model was trained by dataset generated from the validated FEA model. Multiple machine learning algorithms, including OLSLR, Lasso, Ridge, Poisson, RF, KNN, SVR, were applied to predict the first principal stress. By applying four fitting evaluation criteria, it has been found that the nearest neighbor algorithm and the integrated algorithm group performed significantly better than the linear regression algorithm and the support vector machine regression. Among all developed algorithms, the random forest algorithm has the best capability to perform stress prediction for chip package integration design.

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